MAULANA AZAD NATIONAL INATITUTE OF TECHNOLOGY, BHOPAL - 462003

Name of the Program:	B.Tech.	Semester	IV	Year	2nd
Name of the Course:	Power System				
Course Code:	EE 221				
Core/Elective/Other:	Core				
Pre-requisites:	Knowledge of basic electrical engineering and circuit analysis				

Course Outcomes:

1	Understand Electrical power transmission and distribution.
2	Demonstrate overhead transmission lines for power transmission
3	Analyze Performance of transmission lines
4	Evaluate Electrical power distribution systems
5	Design /selection of various parts of a substation.

Description of Content in Brief:

1	General Structure of Power System, Single line diagram, Power Transmission at different voltage level, Type of overhead conductors, Bundle conductors, Skin effect, Proximity effect, Transmission line parameters – Resistance, Inductance and Capacitance calculations – Single-phase and three-phase lines double circuit lines, effect of earth on transmission line capacitance
2	Mechanical design of overhead lines – Line supports, Insulators, Voltage distribution in suspension insulators, string efficiency. Stress and sag calculation – effects of wind and ice loading. Testing of insulators
3	Performance of transmission lines – Regulation and efficiency, Tuned power lines, Power flow through a transmission line, Introduction to Transmission loss and formation of corona, critical voltages, travelling waveform phenomena.
4	Distribution systems – General aspects – Classification of distributed systems (radial/ring and interconnected) design consideration for distributed system, A.C. distribution – Single-phase and three phase. Kelvins law and its limitations
5	Underground cables – Comparison with overhead line, Types of cables, insulation resistance, potential gradient, capacitance of single-core and three-core cables. Layout and design of substation - Design /selection of various parts of a substation

List of Text Books:

1	C.L.Wadhwa, Electric Power System, John Wiley & Sons, 1984
2	D P Kothari and I J Nagrath, Power System Engineering, 3rd Edition, McGraw-Hill
3	~~Asfaq Husain, Electric Power System, 5th edition , CBS, January, 2010~~ A Course in PS by JB Gupta

List of Reference Books:

1	Electrical Transmission and Distribution Reference Book', Westinghouse Electric Corporation, 4thEdition 2007.
2	William D.Stevenson, Elements of Power System Analysis, 4th Revised edition, MGH , 1982
3	TuranGonen, 'Electric Power Distribution System Engineering', CRC Press INC, 2ndEdition 2007.

URLs:

1	https://nptel.ac.in/courses/108/105/108105067/